

**Performance Work Statement
Columbia Mowry O&M New**

1. SECTION C-1: SCOPE OF WORK

The Contractor shall provide all personnel, labor, materials, supplies, equipment (except for a Government-Furnished General Motors Truck with Boom (Section C-2, 2-3)) and supervision necessary to perform the operations, maintenance, and repair activities as required by this Performance Work Statement (PWS) on the following reserve works facilities:

- Columbia Main Pumping Plant
- Columbia Diversion Channel
- Columbia Delivery Canal
- Structures on Columbia Canal
- Columbia Re-lift Pump Plants A, B, C North and C South
- Pipelines located on discharges of Plants A, C North and C South
- Columbia Mowry Pumping Plant and Pipeline

The principal components of work to be performed by the Contractor shall consist of, but are not limited to, the following repair and maintenance activities:

Preventive Maintenance (PM) work and inspections, and corrective actions noted during PM inspections; emergency/repairs and maintenance; and non-emergency/repairs and maintenance.

The operations, maintenance, and repair activities shall be performed on the abovementioned reserve works facilities and any associated feature such as, but not limited to, roads; fences; drainage channels; pumps; signs; gates; bridges; buildings; and all associated mechanical and electrical equipment. Also included is weed, rodent abatement, emergency repairs, minor construction, and other items or services necessary to provide a complete and usable system and perform the work as defined in this PWS.

In the event there is no standard or procedure for a function, it shall be the responsibility of the Contractor to develop such standards and procedures subject to approval by the Contracting Officer (CO) before the proposed standards and procedures are implemented by the Contractor. If the Contractor determines that established standards and procedures should be changed, such changes shall be requested in writing and provided to the CO through the Contracting Officer's Representative (COR). Any changes must receive written approval from the CO before being implemented by the Contractor.

1.1. ADMINISTRATIVE

1.1.1. Administrative Support

The Contractor shall provide the necessary administrative support activities to fully carry out the operation and maintenance program for the reserve works facilities under this contract. This shall include, but not be limited to, the following administrative activities: budget; finance; payroll; environmental quality; safety; personnel management; property management; supply management; maintenance management; and a quality assurance program. All administrative activities shall be consistent with industry standards and performed as required.

1.1.2. Operations and Maintenance (O&M) Manager

The Contractor shall identify an O&M manager who will be responsible for the performance of the work. The name of this person and an alternate who will act for the Contractor when the O&M manager is absent shall be designated in writing to the COR.

The O&M manager or alternate shall have full authority to act for the Contractor on all matters relating to daily operations of this maintenance service contract.

The O&M manager or alternate shall be available during normal business hours within 2 hours of notice to meet with Government personnel designated by the COR to discuss problem areas. After normal business hours the O&M manager or designated alternate shall be available within 4 hours of notice unless a shorter response time is required due to emergency conditions.

The O&M manager and alternate must have a strong command of the English language and be able to communicate both orally and in writing.

1.1.3. Employees

Contractor personnel shall present a professional appearance and be able to provide appropriate identification. This includes the display of proper identification by each employee.

All Contractor and subcontractor personnel shall be in compliance with all Federal and State laws relative to citizenship.

1.1.4. Licenses

Contractor personnel shall have a valid State drivers licenses for the vehicles and/or heavy equipment they are operating. Contractor personnel shall be properly trained to stay current with applicable regulations governing safe equipment operation. Contractor personnel shall comply with all Federal, State, and local laws and regulations regarding pest management and the use of pesticides. The Contractor shall maintain a current list of all Contractor employees and their respective classifications, dates of certifications, and job descriptions

of those certifications.

1.2. REPAIR AND MAINTENANCE ACTIVITIES (Order of Priority)

This paragraph is to establish a schedule of priorities to which the Contractor shall respond and follow in the accomplishment of the specific tasks covered under this contract. Once irrigation season begins, February through November, all delivery related work is on a 24/7 basis. Normal O&M can be done during daylight hours, but the after- hours are on-call, within half-hour response. This half-hour response time is to assure canals are stabilized and deliveries continue while repairs are completed

1.2.1. First Priority

Emergency Work: Continued execution and accomplishment of such work until the immediate emergency is resolved and until further written designation is provided by the COR.

1.2.2. Second Priority

Repair and maintenance items which prevent normal operations and maintenance in progress as either by categorization; as they occur; or as designated by the Contractor and approved by the COR.

1.2.3. Third Priority

Identified in the approved PM schedule. PM Worksheets will be provided to the Contractor by the Government as part of this contract (Technical Exhibit 1).

1.2.4. Fourth Priority

Tasks listed in this PWS, which are not covered by the higher priorities shown above.

1.2.5. Task Prioritization

The Contractor shall maintain and/or repair the reserve works facilities under this agreement using the priorities established above. However, the Government reserves the right of re-establishing the priority schedule when deemed necessary.

1.3. CONTINGENCY PLAN

The Bureau of Reclamation (Reclamation) must plan in advance how it will meet mission requirements in event of a natural disaster, hazardous spill, national emergency, sabotage, or similar incident which would have a significant impact on the ability of the Reclamation reserve works facilities to meet established commitments. Reclamation must be able to react to such events without undue delay. Sudden or unusual events could result in a great impact on the performance and Contract requirements. As an aid to properly plan for such events, the Contractor shall prepare and submit for approval a contingency plan using Emergency Operation Plan for each facility. The Contractor shall outline in detail, the method(s) the Contractor shall use in meeting Contract requirements in the following circumstances:

- (a) A labor strike which impacts the Contractor's ability to perform;
- (b) A hazardous spill which impacts the Contractor to perform, affects public safety, or impairs Reclamation's ability to meet water delivery contractual commitments;
- (c) An incident which causes major damage which impacts the Contractor's ability to perform and
- (d) Other unforeseen circumstances.

In all cases, the Contractor shall assume that Reclamation cannot provide supplemental forces and continues to need the same or additional performance under the Contract, which the Contractor must meet. This plan shall be submitted for review after the award.

1.3.1. Emergency Response

Response to emergency service calls outside of normal business hours shall not exceed four (4) hours regardless of the time during the day or night, weekends or holidays.

Response to emergency service during normal business hours shall not exceed two (2) hours. Response is defined as having all personnel and equipment required to correct an emergency at the site of the emergency or at a pre-approved staging area, ready, willing, and able to perform the emergency work. The Contractor shall provide a draft emergency response plan prior to contract start date.

1.3.2. Emergency Services

On occasion, services may be required to support an activation or exercise of contingency plans outside the normal business hours described above. The Contractor's responsibilities for emergency support shall be in accordance with the contingency plan.

1.4. QUALITY CONTROL

The Contractor shall establish and maintain a complete Quality Control Plan (QCP) to assure the requirements of the Contract are provided as specified. One copy of the

Contractor's QCP shall be provided to the COR. The QCP must be provided to the COR no later than 30 days after the contract award date. The QCP shall be revised in a timely manner whenever changes to the plan occur. The QCP shall include:

- (a) An inspection system covering all the services listed on the Performance Requirements Summary (page 26). It must specify the areas to be inspected on either a scheduled or unscheduled basis, how often inspections will be accomplished, and the title of the individual(s) who will perform the inspection.
- (b) The methods for identifying and preventing deficiencies in the quality of service performed before the level of performance becomes unacceptable or not in conformance with this Contract.
- (c) A system for documenting all inspections conducted by the Contractor and, if necessary, the corrective action taken. The inspection documentation shall be made available to the COR monthly or upon request during the term of the Contract.
- (d) Preventive Maintenance work schedules shall be developed by the Contractor and submitted to the COR. These schedules shall be approved by the COR before being implemented. The CO or designated representative has the option of modifying or redirecting the work if it is determined in the best interest of the Government.
- (e) A written work order status report shall be submitted to the COR by the 10th of each month. This report shall contain information on work accomplished during the previous month, major problems encountered, expected problems, administrative problems, and Maintenance Service Contract management problems.

Following award of the contract and prior to the approval of the QCP and work schedule, the Contractor shall perform work on first and second priority repairs (Paragraph 1.3.1 and 1.3.2) should an emergency occur.

1.5. QUALITY ASSURANCE

The COR will monitor the Contractor's performance under this contract using the quality assurance procedures specified in the Performance Requirements Summary of the PWS (Technical Exhibit 2).

1.5.1. Performance Evaluation Meetings

The O&M manager shall meet at least weekly with the COR during the first month of the contract. Meetings shall be as often as necessary thereafter as determined by the COR. However, a meeting shall be held whenever a Contract Discrepancy Report (CDR) is issued. The Contractor shall provide written minutes of these meetings and the minutes

shall be signed by the COR and O&M manager for concurrence.

1.5.2. Performance Assessment Plan

- (a) **Monitoring Performance.** During the course of the evaluation period, the COR will track Contractor performance. Interim (mid-term) evaluations may be provided to identify strengths and weaknesses in the Contractor's performance during the period being evaluated. At the end of the period, the COR will assess the Contractor's performance in accordance with the Quality Assurance Surveillance Plan (QASP) and report to the CO. Government personnel may observe, photograph, or otherwise record Contract operations. However, these personnel shall not interfere with Contractor performance.
- (b) **Contractor Self-Assessment.** Following each evaluation period, the Contractor may provide a written self-assessment of its performance to the COR to be considered in its report to the CO. The self-assessment shall be submitted not later than 5 working days after the end of each evaluation period. The self-evaluation shall not exceed 1 page per Performance Requirements Summary (PRS) element. The self-assessment shall address both the strengths and weaknesses of the Contractor's performance during the evaluation period. Where deficiencies in performance are noted, the Contractor shall describe the actions planned or taken to correct such deficiencies and avoid their recurrence. The self-assessment itself will NOT be the basis for the payment determination.
- (c) **COR Recommendation.** The COR will consider all evaluations and any other pertinent information, including Contractor self-assessment, and will prepare a report to the CO with findings and recommendations. The Contractor will be provided a copy of the draft findings and recommendations of the COR and will be afforded the opportunity to identify factual errors. The COR's draft recommendation is not subject to negotiation and the COR will not engage in discussions with the Contractor. Any errors identified by the Contractor will be addressed by the COR in its final report. The Contractor will be provided a copy of the final COR report at the same time the report is submitted to the CO.
- (d) **Payment Determination.** The CO may meet with the COR to discuss the COR's report. The Contracting Officer will make a final determination in writing as to the percentage of work successfully completed, and the resulting payment to be made. A copy of the determination will be provided to the Contractor no later than 45 calendar days after the end of the period being evaluated. All CO decisions regarding payment deduction are unilateral decisions made solely at the discretion of the Government.
- (e) **Deduction.** Notwithstanding any other clause of this contract, deduction of the Disincentive will be made within the later of 60 days after the end of the evaluation period or 30 days after receipt of an approved invoice.
- (f) The QASP is one evaluation method the government uses to surveillance

performance to determine whether the Contractor meets the standards of performance as defined in the PWS. The absence of a QASP for any contract requirement, however, shall not detract from its enforceability or limit the rights or remedies of the government under any other provision of the contract in determining the quality of the Contractor performance.

1.6. PHYSICAL SECURITY

The Contractor shall be responsible for safeguarding all Government property provided for Contractor use or under their control. This includes the responsibility of the Contractor to control access to secured areas as well as securing all reserve works facilities after normal business hours.

1.6.1. Key Control

The Contractor shall establish and implement methods of ensuring that all keys issued to the Contractor by the Government are not lost, misplaced, issued to or used by unauthorized personnel. It is the responsibility of the Contractor to prohibit the use of keys issued by the Government by any persons other than the Contractor's employees. The Contractor shall develop procedures covering key control that shall be included in the QCP.

1.7. HOURS OF OPERATION

1.7.1. Normal Business Hours

Maintenance work shall be accomplished during the Contractor's normal working hours. The Contractor's normal working hours shall be from 7:00 AM to 3:30 PM Monday through Friday.

All emergency maintenance or repair work shall be accomplished whenever required and shall be completed without interruption, notwithstanding normally scheduled working hours, weekends or holidays. **All emergency work shall be performed in compliance with Section 1.3.**

1.8. CONSERVATION OF UTILITIES

The Contractor shall minimize the waste of electricity and water. The Contractor shall be directly responsible for instructing employees in utility conservation practices.

Mechanical equipment and control systems should be set in accordance with Federal directives. Hydraulic systems shall be designed and installed to minimize the potential for having an effect on the environment.

When water is no longer needed, water faucets or valves shall be turned off. All leaking faucets, and valves, etc., shall be repaired promptly.

1.9. SAFETY REQUIREMENTS

Department of Labor Code of Federal Regulations Title 29 and other Federal, State and local government safety standards shall apply in its entirety to the Contractor. Examples of applicable safety standards are OSHA 29 CFR 1910, OSHA 29 CFR 1926, California Title 19, "Public Safety", California Title 8, "General Industry Safety Orders", and 41 CFR, Subpart B, "Public Contracts, General Safety and Health Standards." In addition to the above, the Contractor should also use the "RECLAMATION SAFETY AND HEALTH STANDARDS" for reference (The yellow book).

1.9.1. Safety of the Public

Roads, subject to interference by the work, shall be kept open or suitable detours shall be provided and maintained by the Contractor. The Contractor shall provide, erect, and maintain all necessary barricades, suitable and sufficient flasher lights, flaggers, danger signals, signs, and shall take all necessary precautions of the work and the safety of the public. Such work shall be done in accordance with all applicable State, Federal, and local regulations.

Roads closed to traffic shall be protected by effective barricades on which shall be placed acceptable warning and detour signs. All barricades and obstructions shall be illuminated at night, and all lights shall be lighted from sunset until sunrise.

Specific signs, barricades, and flagmen requirements are detailed in the American National Standards Institute "Manual on Uniform Traffic Control Devices for Streets and Highways" (ANSI D6.1). The Contractor must conform to these standards and all other applicable Federal, State and County regulations.

1.9.2. Safety and Health

The Contractor is responsible for being cognizant of and insuring compliance with the following requirements:

- (a) The Contractor shall not require, or allow any employee, engaged in the performance of this Contract to work under conditions which are unsanitary, hazardous, or dangerous to their health or safety, as determined under safety and health standards promulgated by the Secretary of Labor under Section 107 of the Contract Work Hours and Safety Standards Act (40 U.S.C. 327 et seq.) as amended and other Federal, State and applicable local government safety standards. The worker is ultimately responsible for their own safety and shall not be required to perform work they believe to be unsafe.

- (b) The Contractor shall fully comply with Federal, State and local government safety and health standards and amendments or revisions thereto in effect on the date of the contract. Federal Standards are available from the Government Printing Office in Washington D.C. Construction Safety and Health Standards promulgated by the Secretary of Labor may be obtained from any regional or area office of the Occupational Safety and Health Administrative of the U.S. Department of Labor. State of California Standards are available through Barclays Law Publishers, San Francisco, CA.

Such responsibility shall apply to both the Contractor operations and those of their subcontractors. When violations of the safety and health requirements contained in these specifications of standards referenced in Subparagraph 1.9.2.a and 1.9.2.b are called to their attention by the COR or designated representatives, the Contractor shall immediately correct the conditions to which attention has been directed. Such notice, either verbal or written, when served on the Contractor or designated representative(s), shall be deemed sufficient. In the event the Contractor fails or refuses to promptly comply with the compliance directive issued, the CO or designated representative may issue a written order to stop all or any part of the work. When satisfactory corrective action is taken, a written order to resume work will be issued. The Contractor shall not be entitled to any extension of time, or to any claim for damage or to additional compensation by reason of either the directive or the stop order. Failure of the CO or designated representative to order discontinuance of any or all of the Contractor's operations will not relieve the Contractor of his responsibility for the safety of personnel and property. The rights and remedies of the Government provided in this paragraph are in addition to any other rights and remedies provided by law or under this contract.

1.9.3. Safety Program

The Contractor shall submit, 14 days after award, in writing a proposed safety program as required in CAL-OSHA Standards, as prescribed in SB 198, Chapter 1359, Labor Code 6401.7 based on anticipated operations, maintenance, and construction activities as related to safety and health standards.

In the event there is a conflict between the requirements contained in the Contractor's approved safety program, governmental safety and health codes, and standards, including the U.S. Department of Labor Construction Safety and Health Standards, promulgated under Section 107 of the Contract Work Hours and Safety Standards Act (40 U.S.C. 327 et seq.), as amended, the more stringent requirement will prevail.

1.9.4. Safety Records

The Contractor shall maintain an accurate record of, and shall report to the COR in the manner prescribed by the COR, all cases of death, lost-time accidents, vehicle accidents, occupational diseases, or traumatic injury to employees or the public involved, and any property or equipment damage incident to the performance of work under this contract.

1.10. PUBLIC CONTACT

Where possible, Contractor personnel should develop positive working relationships with adjacent landowners within their area of responsibility.

Contractor personnel shall inform any trespassers along the right-of-way, of hazards and when necessary request trespassers to leave restricted areas immediately.

When Contractor personnel are confronted with media inquiries, the O&M manager shall refer all media inquiries to the COR.

When Contractor personnel encounter emergency personnel, police or firefighters along the right-of-way responding to an emergency, Contractor personnel shall notify the O&M manager. The O&M manager shall keep the COR informed of all emergencies occurring along the right-of-way and/or on Reclamation reserve works facilities.

1.11. MODIFICATIONS TO EXISTING EQUIPMENT

Any modifications made to existing equipment that will affect the operation and/or control of the equipment being modified, require changes or revisions be made to existing drawings, or require generation of a new drawing will need prior approval from the COR.

1.12. HAZARDOUS MATERIALS AND HAZARDOUS RESPONSE

The Contractor shall conform to all Reclamation, Federal, State, and local regulations in the use and disposal of hazardous materials and hazardous waste. The Contractor shall provide the appropriate response to any emergency involving hazardous materials at any of the reserve works facilities covered by this contract. The Contractor shall provide proof/certification of its personnel qualified to work with hazardous materials and wastes to Reclamation. The Contractor shall notify Reclamation immediately upon discovering a release of hazardous waste into the environment.

Examples of Federal regulations on hazardous materials are found in the following:

1. 40 CFR 100-399 (Water Quality Programs, Pesticides and Hazardous Waste)
2. OSHA 29 CFR 1910.120 (Hazardous Waste Operations and Emergency Response); 29 CFR 1910.120 (Hazard Communication)
3. 49 CFR 172 (Subpart B – Tables of Hazardous Materials)

Examples of State regulations on hazardous materials are:

1. California Title 22 (Environmental Health)
2. California Title 26 (Toxics)

SECTION C-2

2. SECTION C-2: SERVICES

This section describes the Services that shall be required as part of this contract.

2.1. TRAINING

The Government will provide Contractor personnel onsite training in the applicable Government materials and regulations as required by the COR. This includes methods for obtaining supplies in accordance with Government procedures. This training shall be provided by the Government as determined necessary by the Contractor or by the Government.

2.2. GOVERNMENT SURPLUS MATERIALS AND EQUIPMENT

The Contractor shall be authorized to screen government surplus materials and equipment for use in support of the scope of this contract. The COR will review and approve/disapprove all requests for the transfer of government surplus materials and equipment to the Contractor.

2.3. GOVERNMENT-FURNISHED EQUIPMENT

The Government will provide furnished equipment for Operations and Maintenance of the Columbia Mowry Facilities under the provisions of FAR 45.000 to provide and use Government-Furnished property in the contract. Specific equipment includes, but not limited to, PCN 015191, License 1471583, Truck, General Motors, 8-cylinder, gas, 4x2, with boom, 2.5 ton, 34,000 GVW (Truck).

No major repairs will be made to the Truck without approval of the COR. Upon completion of Contract for Operations and Maintenance, the Truck will be returned to the Government in a like condition with consideration given for normal wear and tear. Monthly usage and mileage (trip sheet) will be reported monthly to the Government to reflect mileage, operational hours used, gasoline, and repairs and/or maintenance costs.

SECTION C-3

3. SECTION C-3: GENERAL TASKS and SPECIFIC WORK

3.1. GOVERNMENT LIAISON

The COR shall act as the liaison between the Reclamation and other Federal, State, and local agencies, private individuals, and or entities for planning and related activities which affect the reserve works facilities designated in this contract. The Contractor activities shall include the following:

Review and comment on planning documents, including preliminary and final designs, preliminary and final development plans, environmental documents, and other documents relating to Federal, State, and local planning procedures. A copy of the Contractor's comments shall be forwarded to the COR, participate as appropriate with Federal, State, and local agencies, or private individuals and entities. The Contractor shall keep the COR informed of these activities and prepare such emergency preparedness plans and other plans as may be required by various Government agencies. The Contractor shall furnish copies of any such plans to the COR.

3.2. MAINTENANCE RECORDS

Any maintenance performed by government forces on equipment or reserve works facilities described under this contract will be documented on the appropriate Preventive Maintenance worksheet.

3.3. PEST MANAGEMENT

Pest management activities performed on, in, or around all lands, reserve works facilities and appurtenant structures designated in this contract shall include but are not limited to the following: Terrestrial weed control, Aquatic pest control, Rodent control, and Insect and invertebrate control.

Johnson grass (*Sorghum halepense*), Russian knapweed (*Centraurea repens*), groundsel (*Senecio vulgaris*), redstem filaree (*Erodium cicutarium*), Russian thistle (*Salsola pestifer*), Bermuda grass (*Cynodon dactylon*), and turkey mullein (*Eremocarpus setigerus*), are but a few of the commonly encountered noxious and troublesome agricultural weed pests.

Filamentous green algae (*Cladophora* spp., *Ulothrix* sp., *Oedogonium* spp., etc.), blue-green algae (*Anabaena* spp., *Phormidium* spp., etc.), American pondweed (*Potamogeton nodosus*), waterweed (*Elodea canadensis*) and Asiatic clam (*Corbicula fluminea*) are but a few of the commonly encountered and edemic aquatic pests.

Ground squirrels (*Spermophilus beecheyi* and *Spermophilus beldingi*) are some, but not all, of the most commonly encountered and troublesome vertebrate (rodent) pests.

Black widow spiders (*Lactrodectus mactans*), yellowjacket wasps (*Vespula maculifrons*), and mud-dauber wasps (*Sceliphron caementarium*) are three examples of some of the more

bothersome insects and invertebrates.

When deemed appropriate, necessary and/or in keeping with CO objectives, all lands, reserve works facilities and associated features designated in this contract shall be subject to some form of pest management. The types of detailed features requiring some form of pest management consist of but are not limited to the following: maintenance and operations roads, canal banks, levees, ditches, right-of-way areas, maintenance and equipment yards, borrow areas, fence lines, building perimeters and sumps.

During all phases and sub-tasks of the pest management program, all applicable Federal, State and local laws, regulations, ordinances and guidelines shall be followed by the Contractor. These phases and sub-tasks include the following (not an exhaustive list):

Procurement, transportation and storage of pest control materials; Planning, documentation, registration and certification procedures for any and all pest control activities; Proper and applicable safety procedures and equipment; Initiation, application and monitoring of pest control activities, materials, and their residues.

3.4. TERRESTRIAL WEED CONTROL

All terrestrial weed control activities by the Contractor shall be compatible and complementary with the onsite and adjacent land uses and cultural practices. Overall, these activities will provide or help provide for a safe working environment in, on, and around all reserve works facilities designated in this contract. In addition, these activities will minimize (or exclude whenever possible) any negative impacts on the efficient operation and maintenance of reserve works facilities. The Contractor shall perform weed control on the reserve works facilities designated in this Contract based on the intended or present land use.

3.4.1. Joints and Cracks

Joints and cracks on all structures designated in this contract shall be kept vegetation-free by the Contractor.

3.4.2. Channel Banks

The channel banks shall, whenever practicable, be maintained by the Contractor in such a way that noxious and troublesome agricultural weeds are controlled or excluded. A complex or monoculture of grasses and/or herbaceous plants compatible to adjoining land uses is to be left for purposes of erosion control. When this primary goal/standard cannot be effectuated for reasons such as, but not limited to climate, soil type, or weed diversity, then all noxious and troublesome agricultural weeds shall be controlled or excluded. The weed pests shall be temporarily or permanently excluded from propagating and proliferating on said site. The weed pests shall be controlled in such a manner that they do not:

Significantly hinder, foul, plug, destroy, encroach upon, cover, or interfere with any

Reclamation reserve works facilities and; Impact, in any deleterious, competitive, or harmful manner on the adjoining land uses.

3.4.3. Roads

All roads shall be maintained by the Contractor in a vegetation-free state.

3.4.4. Right-of-Ways

The Contractor shall maintain Reclamation right-of-way in such a manner that all noxious and troublesome agricultural pests are controlled or excluded.

3.4.5. Areas Where Vegetation is Excluded

Maintenance yards, buildings and facility perimeters, parking lots, and any other areas designated in this Contract that require the total exclusion of vegetation shall be maintained in such a state by the Contractor.

3.5. AQUATIC WEED CONTROL

The water conveyance facilities designated in this contract shall be maintained by the Contractor free of algae, aquatic weeds and other aquatic pests to such an extent that the flow capacities, applicable water quality, design capabilities and characteristics, operations, and maintenance of reserve works facilities are not infringed upon nor adversely impacted. The aquatic pest shall also be maintained so that said pests do not adversely affect the water users and recipients, and other Parties and entities that receive the services and goods provided by Reclamation.

The aquatic pest control activities shall be in accordance with, and compatible to, all various uses for which the water has been contracted and been made available.

3.6. RODENT CONTROL

Rodent populations on all lands and reserve works facilities designated in this contract shall be controlled by the Contractor at population levels that do not pose a safety hazard to the O&M work force, Reclamation reserve works facilities and the general public. In addition, the rodent populations shall be reduced to levels that do not:

Interfere with any other operation and maintenance functions; significantly degrade or threaten the structural integrity and physical appearance of the facility; degrade the safety and structural integrity of all operations and maintenance roads; and serve as a "core" population that could infest and establish other populations on adjacent lands.

3.7. INSECT CONTROL

The Contractor shall minimize the insect and invertebrate pests at the reserve works facilities designated in this contract that pose a nuisance of safety hazard to individuals

temporarily occupying or working in and around those structures or; have the potential to destroy, degrade, infest, impair, or adversely alter the inherent integrity, consistency, utility, and effectiveness of any material comprising the structures themselves, or contained within, on, or around said structures.

3.8. RIPRAP

Riprapping for erosion protection and control may be required on any of the structures associated with the reserve works facilities designated in this contract. These structures include, but are not limited to; channel embankments, bridge abutments, gate structures and drainage structures.

3.8.1. Riprap Inspection and Repair

The Contractor shall provide routine inspection of channel and drainage systems and other pertinent reserve works facilities for need of additional riprap or repair of existing riprap as per requirements of the Government.

When voids occur in the riprap, the Contractor shall fill these voids.

The principal components of the maintenance work performed on the riprap consist of but are not limited to the following:

Grading, slide repair, placement of riprap, hauling and excavation.

3.9. PAVED AND UNPAVED ROADS

The Contractor shall be required to maintain two types of operating and maintenance roadways at reserve works facilities designated in this contract. The types of roadways are paved roadways and unpaved (dirt) roadways.

The principal components of the maintenance work performed on the roadways consist of the following work, but not limited to: blading; crack sealing; filling potholes; dust control; sweeping; weed control; watering; sloping; surfacing; sealing; and material hauling.

3.9.1. Paved Road Standards

- (a) The Contractor shall repair all roadway potholes prior to the rainy season.
- (b) After the rainy season, the Contractor shall repair damaged road areas in a manner which allows safe passage on the roadway for any maintenance or inspection activities.
- (c) During the rainy season, the Contractor shall keep the road passable for operations and maintenance vehicles and equipment.

- (d) All cracks that develop a gap wider than 1/2" shall be sealed with a Reclamation approved crack sealant.
- (e) The Contractor shall resurface the asphalt road surfaces on an as needed basis or as directed by the COR.

3.9.2. Unpaved (Dirt) Roads Standards

The roads shall be maintained passable for operation and maintenance equipment and free of weeds and grasses.

3.9.3. Signs and Reflectors

All signs are installed to provide warning, instruction, location and direction.

Principal components of the maintenance work performed on signs consist of the following work, but not limited to: installing, repairing, replacing, and painting.

The Contractor shall inspect all signs annually and repair, replace or paint as required. The Contractor shall maintain painted signs, so they are bright and readable from 100 feet.

3.10. GATES

Gates can vary from standard chain-link with 3-strand barbed wire top, double opening, pipe gates, chain link gates to standard barbed wire, wood post stock fence gates, Texas gates, wood gates, etc. They shall vary in height and width depending on need.

The Contractor shall annually inspect and replace, repair, paint and lubricate as required to keep gates in good operating condition.

3.10.1. Typical Maintenance

The principal components of the maintenance work performed on gates consist of, but are not limited to, the following work:

Adjusting swing; repairing; replacing; painting; repairing or replacing reflectors; repairing or replacing locking system; lubricating; and painting.

Gates shall be replaced, repaired and maintained by the Contractor to Bureau standards. The Contractor may contact the COR to request these standards.

3.10.2. Access to the Columbia Mowry Facility

Typically, gates are located at the entrance to reserve works facilities. The exception is the Columbia Mowry Facilities where the facilities are first accessed through a private, automatic opening gates, located east of the pumping facilities. Reclamation will be notified of any change in the combination currently in use. Once at the pumping facility site, the

gates are a standard chain-link gate.

3.11. FENCES

This covers all fencing surrounding work areas which include system structures and maintenance yards. Types of fences in place in this contract are barbed wire, 4-strand or 5-strand with steel and wood posts and chain link ranging from a height of 4 feet to 8 feet with 3-strand barbed wire top.

The Contractor shall inspect fences around the reserve works facilities monthly and repair, replace and paint as required.

3.11.1. Typical Maintenance

The principal components of the maintenance work performed on fences consist of, but are not limited to, the following work:

Repair; replacement; removal; and painting.

Fences shall be replaced, repaired and maintained by the Contractor to Bureau standards. The Contractor may contact the COR to request these standards.

3.12. GUARDRAIL

Guardrails are used to provide protection to structures, to equipment, to keep vehicles out of the river or channels and for employee safety. Typical locations are where roads cross the river or channels where a road dead-ends into the river or drain and where operating roads go by structures or stationary equipment. Guardrail may be made of the following materials and combinations of pipe, wood and metal.

The Contractor shall inspect annually and repair, replace, and paint reflectors as required.

3.12.1. Typical Maintenance

The principal components of the maintenance work performed on guardrail consist of the following work but is not limited to:

Repairing, replacing, painting, and constructing as requested.

3.13. ELECTRICAL MAINTENANCE

3.13.1. General

The main electrical components for the electrical maintenance of reserve works facilities consist, of but are not limited to, the following: new construction; maintenance; repairing; conduit, single-phase and three-phase motors; switches; control panels; secondary contacts and wiring; batteries: power and control circuits; lighting systems

To assure high quality service, the Contractor shall have a definite maintenance schedule, including preventive maintenance, systematically followed. Operating and local conditions vary to such an extent that the schedule shall be prepared to suit the conditions. However, the maintenance schedule for individual devices, such as circuit breakers, relays, batteries, meters, etc., shall be based upon current industry standards.

To safeguard against equipment failure, the electricians and technicians will regularly service the electrical equipment. The frequency of inspections is a variable, but this program shall be planned and carried out as scheduled. This is a basic requirement of any maintenance program which is aimed at preventing unscheduled outages.

3.13.2. Cleanliness

Dirt is a principal cause of electrical failure; it's a day-to-day accumulation of normal atmospheric particles, lint, chemicals, or metallic particles from associated motors, oil mist, and vapors. These deposits, if allowed to accumulate, will contaminate electrical equipment and cause failures. During the performance of routine preventive maintenance, the Contractor shall keep all electrical equipment clean.

The Contractor shall maintain electrical machinery as clean as practical.

3.13.3. Dryness

Electrical apparatus operates best in a dry atmosphere for several reasons. Humidity can cause copper, aluminum, iron, or alloyed parts to oxidize. A buildup of oxides, if caused by caustic fumes, or acids, can destroy metal and affect electrical connections and contacts. Heavy accumulation of oxides shall be removed and severely corroded parts replaced. High humidity results in free moisture condensing on the equipment. This can cause short circuiting and immediate failure. The Contractor shall control humidity, if possible, because moisture can increase dirt buildup on electrical parts; this can lead to failure. For humid applications, the safest course is to use equipment designed to operate under high moisture conditions. In most cases, this shall require waterproof enclosures.

The Contractor shall keep all electrical apparatus as dry as practical.

3.13.4. Tightness

Certain components of electrical equipment operate with high-speed movement. This is particularly true of contactors or other control devices, which cause wear of moving parts and other control devices. Appreciable imbalance tends to create vibrations in the equipment and loosen hardware and other parts. Imbalance is frequently increased by external vibrations set up by unrelated machinery. Continued vigilance shall be necessary to deter wear and loosening of apparatus parts or connections. Routine servicing shall include a check for tightness of accessible hardware and bolted parts as a simple precautionary measure. On rotating apparatus, vibration can cause mechanical damage and if allowed to

become excessive, can affect the alignment and cause bearing failures, loosening of coils or support generally follows. Any electrical apparatus using bearings is vulnerable in this respect.

The Contractor shall keep all electrical apparatus connections as tight as possible.

3.13.5. Friction Prevention

The Contractor shall promote the prevention of friction in all electrical devices

3.13.6. Buses and Connections

The Contractor shall inspect for systems that indicate overheating or weakened insulation.

3.13.7. Meter Readings

The Contractor shall maintain and track all meter readings

3.13.8. Main Disconnect Contacts and Supports

Each breaker is removed from its housing and all circuits and exposed primary contacts and their supports are de-energized. The Contractor shall inspect for abnormal wear or overheating and wipe clean with the proper solvent recommended by the manufacturer. If discoloration is noted on the surfaces, it is not harmful, unless corrosion due to atmospheric conditions is severe, leaving deposits on the surfaces. All deposits shall be removed by rubbing with a clean cloth (don't use emery cloth or file). If the contacts are severely burned, new contacts shall be installed. Each breaker is checked (service while it is out of the housing) and if need be, replace breakers with the type recommended.

3.13.8.1. Secondary Disconnecting Contacts

All secondary disconnecting contacts shall be inspected for abnormal wear, fatigue, or overheating. Contacts will be replaced if necessary, otherwise they are treated the same as main disconnecting contacts.

3.13.8.2. Secondary Wiring Connections

All wiring connections shall be checked for tightness and at terminal blocks where circuits leave the switchgear. The Contractor shall make sure that all secondary wiring connections are properly connected to the switchgear ground bus where indicated. Look for broken wire strands and pinched or damaged insulation.

3.13.9. Instruments, Relays and Relay Mounted Device

Individual devices shall be maintained by the Contractor according to the specified instructions supplied for each device. All relay covers are to be removed and the interior inspected for dust or dirt. The contacts on control switches, transfer switches, and instrument switches, shall be inspected by the Contractor.

All instruments, relays and relay-mounted devices shall be inspected for proper operation during routine maintenance procedures and repair or replaced when necessary.

3.13.10. Ventilation

The air space under outdoor switchgears is necessary for the entrance of ventilating air; it must be clear of leaves, debris, or obstruction.

All labyrinths, grillwork, and air passages shall be checked for obstructions and accumulation of dirt.

3.13.11. Lubrication

In general, there is nothing that requires lubrication in the stationary housings. However, the main disconnecting finger and stationary contacts are to be cleaned periodically and kept dry. Vaseline or grease shall not be applied to any of the contacts surface.

The lubrication of the draw-out breakers are to be maintained in accordance with the manufacturer's instruction manual.

3.13.12. Renewable Parts

When parts are ordered, all necessary information shall be included, all information can be found on the nameplate, or from equipment records. Parts may consist of: bearings, motors, pumps, valves, and seals.

3.13.13. General Scheduled Inspections and Maintenance

The switchgear installation shall be given a thorough cleaning and maintenance check twice a year. This shall be done at the beginning of the water year and towards the middle of the water delivery season.

3.14. OPERATIONS AND MAINTENANCE OF THE COLUMBIA MOWRY DISTRIBUTION FACILITY

General the Contractor shall operate pumps at various locations as described below. Pumps shall be maintained in good working order so as not to cause an interruption in service. The Contractor is responsible for the following:

3.14.1. Start-Up of Pumps

The following is to be inspected prior to initial start-up each year: Electrical panels, clean out spiders, wasps, etc., check fuses and wires and replace where needed; and oil units as needed.

3.14.2. Daily Maintenance Requirements

Checking oil drippers for leaks and excessive flow; maintaining all oil supply tanks-at least 80% full at start-up, refill as necessary, cleaning all trash racks, inspecting oil dripper units at each pump for proper operation and leaks, keeping area inside pump station clean of debris and oil, maintaining all security gates, signs, etc., repair/replace as required and completing and filling out maintenance logs.

3.14.3. Pumping Plant "C" North, "C" South

Locking off power to pumps during non-delivery periods, "C" North discharge ditch should be inspected daily when operating; flap valve needs to be maintained in good working order. Since "C" South distribution line is underground and subject to farm implement damage, the Contractor should inspect line frequently for breaks or damage.

The special notices regarding the operation of Pumping Plant "C" North, consist of, but are not limited to, the following:

- (a) Automatic liquid level control sensors are set and not to be adjusted;
- (b) Automatic liquid level control sensors should be lubricated as often as necessary to assure smooth movement of the probe;
- (c) Water requests will be administered by Contractor staff only;
- (d) Chain - Emergency offs should be reported to ditch operators as soon as possible; and
- (e) All farm operators should have contact cards for the Contractor.

3.14.4. Pumping Plant 'B'

Cleaning Inlet trash racks, maintaining upstream float boards in good working order; reporting leaks of underground pipelines that discharge to the south, preventing access to power panel when not in operation, maintaining oil levels in supply tank and pump site glass.

3.14.5. Pumping Plant 'A'

Daily cleaning of trash rack, maintaining float boards, and replacement when needed, units one and two must be inspected at least once during irrigation season to determine that a pump in the off position is not turning in reverse while the other pump is running, inspecting discharge line underground to the west ½ mile, then north one mile – for damage to line and turnouts as required, keeping fence enclosure secure at all times.

3.14.6. Measuring Walkway and Recorder Buildings

One galvanized walkway is installed across the concrete delivery canal to the north of the main plant. One concrete block house is used by Columbia Canal Company to house a water quality meter and recorder.

Maintaining the above walkway and block recorder in good working order.

3.14.7. Main Pumping Plant and Seepage Pump

Reporting, immediately any unusual sounds, vibrations, oil leaks or any electrical panel problems, cleaning trash racks to keep them free of debris, keeping clean flow meters and hour clocks and reporting problems with the equipment, daily inspections of oil dripper line for leaks and adequate flow, maintaining security for all gates, locks, fences, and railings, and keeping the public out of this area. Should there be problems, contact the O&M manager and or COR for assistance.

Groundwater and surface water levels on the former MLT property shall be managed to maintain dry road and access conditions to the Columbia Main Pumping Plant and diversion facilities as well as to prevent surface water ponding in the adjacent fields.

3.14.8. Mowry Pumping Plant

Servicing and maintaining all equipment while deliveries are being made, conducting daily checks, inspecting underground delivery line – located to the south – for leaks, maintaining Mowry Canal flap gate in proper operating condition, area around meter pit to be kept dry and maintained, maintaining inlet from San Joaquin River by keeping it weed free, keeping brush away from power poles at rivers edge, and rip-rap when necessary.

3.14.9. Safety Equipment and Procedures

Maintain all walkways and handrails; Inspect bridges for wear, damage, or needed repairs; Maintain vehicles in safe working order; Inspect all pump facilities for damage, vandalism, etc.; Repair/replace any unsafe condition; conduct safety meetings minimum of 4 times per year. Report all injuries to Manager as soon as possible.

Security measures at all reserve works facilities: Facilities inspected daily for vandalism or any other security problems.

3.14.10. Meter Reading and Quality Control Monitoring Program

The Contractor shall develop a Standing Operating Procedure (SOP) for the reading of the water meters and shall include, but not be limited to, procedures for the calibration and testing of meters (devices), testing schedule, written testing procedures, a listing of showing location of devices, rotation program for the meter readers to facilitate detection of potential irregularities, a record keeping program and formal documentation of procedures.

3.15. WATER OPERATIONS OF THE COLUMBIA MOWRY DISTRIBUTION FACILITY

The Contractor shall operate the distribution system to ensure water deliveries to customers at various locations. Pumps shall be maintained in good working order so as not to cause an interruption in service. The Contractor is responsible for the following:

3.15.1. Irrigation Season

Once irrigation season begins, February through November, the distribution system is manually operated, and all delivery related work is on a 24 hour and 7 day a week basis.

3.15.2. Water Orders and Water Quality Control Monitoring Program

During irrigation season, water orders are received daily by the Ditch Tenders. Ditch Tenders shall keep daily records of water orders and water flow rate measurements along entire distribution system (weir sticks, flow meters, etc.) for quality control. Water deliveries may vary from customer to customer depending on their water needs.

3.15.3. Regulating the Distribution System

After daily water orders are received, Ditch Tenders will regulate the distribution system daily to meet water orders demands approximately at 8:00 a.m. or as needed by adjusting dial settings for all pumping plants pumps. Adjust wooden board checks along the entire open waterways to ensure proper hydraulic head for water delivers.

3.15.4. Required Personnel

The Contractor shall have at least 2 Ditch Tenders available for water operations with the following criteria's:

3.15.4.1. Primary Ditch Tender

Primary Ditch Tender must have at least 10 years of water distribution system with pumping plant experience.

3.15.4.2. Secondary Ditch Tender

Secondary Ditch Tender must have at least 5 years of water distribution system with pumping plant experience.

3.15.4.3. Additional Ditch Tenders

Any additional Ditch Tender must have at least 5 years of water distribution system with pumping plant experience.

**COLUMBIA-MOWRY FACILITIES PREVENTATIVE
MAINTENANCE SCHEDULE (5-YEAR PLAN)**

4. SECTION C-4: TECHNICAL EXHIBITS

4.1. Preventative Maintenance Schedule

Facility	2026	2027	2028	2029	2030
'C' North		Pull/check		Replace Trash Racks	Pull/check
'C' South	Pull/check		Pull/check		Replace Trash Racks
'B" Plant		Pull/check		Replace Trash Racks	Pull/check
'A" Plant					
Unit #1	Replace Trash Racks		Pull/check		Pull/check
Unit #2	Replace Trash Racks		Pull/check		Pull/check
Main Plant					
Unit #1	Replace Pump Gates	Replace Trash Racks	General Repairs	Replace Flap Gates	General Repairs
Unit #2	Replace Pump Gates	Replace Trash Racks	General Repairs	Replace Flap Gates	General Repairs
Unit #3	General Repairs	Replace Pump Gates	Replace Trash Racks	Replace Trash Racks	Replace Flap Gates
Unit #4	General Repairs	Replace Pump Gates	Replace Trash Racks	Replace Trash Racks	Replace Flap Gates
Unit #5	Replace Flap Gates	General Repairs	Replace Pump Gates	Replace Trash Racks	Replace Pump Gates
Unit #6	Replace Flap Gates	General Repairs	Replace Pump Gates	Replace Trash Racks	Replace Pump Gates
Mowry					
Unit #1	Pull/check		Pull/check	Repair Bearing	
Unit #2	Repair Bearing		Pull/check		Pull/check
Diversion Channel			Excavate Canal		

Delivery Canal Roads	Add gravel west side	Add gravel west side	Add gravel west side	Add gravel west side	Add gravel west side
Repaint Facilities as needed	Repaint as needed	Repaint as needed	Repaint as needed	Repaint as needed	Repaint as needed

4.2. Performance Requirements

Attachment 1a - PWS - Technical Exhibit 1 - QASP

Performance Requirements Summary For O&M of Columbia Mowry Distribution Facilities

Performance Requirements Summary	Performance Standard	Acceptable Quality Level	Method of Surveillance
1. Operations and Maintenance			
1.1 Emergency Repairs	Timely Response, Timely Completion, Quality Work, Proper Procedures	100% Timely Response and Completion, Quality Work: Comply with industry standard	100% Inspection - Each month, the COR, shall review the generated documentation and enter summary results of all work completed by the contractor into the Surveillance Activity Checklist. Periodic Inspection - The COR will typically perform the periodic inspection on a monthly basis.

1.2 Preventive Maintenance	(FIST) 100% of the time. Proper Procedures: Industry Standard and Operation Manual are followed 100% of time.	Random Inspection
1.3 Repairs & Maintenance Items		100% or Periodic Inspection
1.4 Deferred Maintenance, New Construction and Upgrades		Periodic Surveillance

PERFORMANCE REQUIREMENTS FOR O&M OF COLUMBIA MOWRY FACILITIES

Required Task	Standard	Acceptable Quality	Surveillance	Date of Inspection	Comments
2.1 Government Liaison					
2.2 Maintenance Records					
2.3 Pest Management					
2.4 Terrestrial Weed Control					

2.5 Aquatic Weed Control					
2.6 Rodent Control					
2.7 Insect Control					
2.8 Riprap					
2.9 Berms and Ditches					
2.10 Paved and Unpaved Roads					
2.10.1 Signs/Reflectors					
2.11 Gates					
2.12 Fence Repairs					
2.13 Guardrails					

PERFORMANCE REQUIREMENTS FOR O&M OF COLUMBIA MOWRY FACILITIES

Required Task	Standard	Acceptable Quality	Surveillance	Date of Inspection	Comments
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2.14 Electrical Maintenance					
2.14.2 Cleanliness					
2.14.3 Dryness					
2.14.4 Tightness					
2.14.5 Friction Prevention					
2.14.6 Buses & Connections					
2.14.8 Main Disconnect Contracts & Supports					
2.14.9 Instruments & Relay Mounted Devices					

PERFORMANCE REQUIREMENTS FOR O&M OF COLUMBIA MOWRY FACILITIES					
Required Task	Standard	Acceptable Quality	Surveillance	Date of Inspection	Comments
2.15.2 Daily Maintenance Requirements					

2.15.3 Pumping Plant C-North C-South					
2.1.5 Pumping Plant 'A'					
2.3.11 Gates					
2.3.12 Fences					